

WEB DEVELOPMENT

REVIEW -2

CALORIES CALCULATOR PROJECT REPORT

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CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Health Dashboard</title>

<script
src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
<style>
```

```
*{margin:0;padding:0;box-sizing:border-box;font-family:'Segoe UI',sans-serif;}
```

```
body{
```

```
    min-height:100vh;
```

```
    background:linear-gradient(135deg,#BFBBF8,#667eea,#764ba2);
```

```
    display:flex;
```

```
    justify-content:center;
```

```
    align-items:center;
```

```
    color:#fff;
```

```
}
```

```
.card{
```

```
    width:420px;
```

```
    background:rgba(255,255,255,0.15);
```

```
    backdrop-filter:blur(20px);
```

```
    padding:30px;
```

```
border-radius:20px;  
}
```

```
input,select{  
    width:100%;  
    padding:10px;  
    margin:8px 0;  
    border:none;  
    border-radius:10px;  
}
```

```
button{  
    width:100%;  
    padding:12px;  
    border:none;  
    border-radius:20px;  
    background:linear-gradient(45deg,#00f2fe,#4facfe);  
    font-weight:bold;  
    cursor:pointer;  
    margin-top:10px;
```

```
}
```

```
.dashboard{display:none;}
```

```
canvas{margin-top:20px;}
```

```
.password-box{position:relative;}
```

```
.eye{
```

```
    position:absolute;
```

```
    right:10px;
```

```
    top:50%;
```

```
    transform:translateY(-50%);
```

```
    cursor:pointer;
```

```
    color:black;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- LOGIN -->
```

```
<div class="card" id="authCard">
```

```
<h2>Login</h2>
```

```
<input type="email" id="email"  
placeholder="Email">
```

```
<div class="password-box">
```

```
<input type="password" id="password"  
placeholder="Password">
```

```
<span class="eye" id="eyeIcon"  
onclick="togglePassword()">  </span>
```

```
</div>
```

```
<button onclick="login()">Login</button>
```

```
</div>
```

```
<!-- DASHBOARD -->
```

```
<div class="card dashboard" id="dashboard">
```

```
<h3>Welcome, <span id="userEmail"></span> 
```

```
</h3>
```

<input type="number" id="age" placeholder="Age">

<input type="number" id="height"
placeholder="Height (cm)">

<input type="number" id="weight"
placeholder="Weight (kg)">

<select id="gender">

<option value="">Gender</option>

<option value="female">Female</option>

<option value="male">Male</option>

</select>

<h4>🍴 Select Meals</h4>

<label><input type="checkbox"
onclick="toggleMeal('breakfastBox')">
Breakfast</label>

<div id="breakfastBox" style="display:none;">

<label><input type="checkbox" value="140"
onchange="updateMealCalories()"> 2 Idlis(140
cal)</label>

<label><input type="checkbox" value="360"
onchange="updateMealCalories()"> 3 Dosa(360
cal)</label>

<label><input type="checkbox" value="180"
onchange="updateMealCalories()"> Pongal(180
cal)</label>

<label><input type="checkbox" value="170"
onchange="updateMealCalories()">Oots 1cup(170
cal)</label>

</div>

<label><input type="checkbox"
onclick="toggleMeal('lunchBox')"> Lunch</label>

<div id="lunchBox" style="display:none;">

<label><input type="checkbox" value="250"
onchange="updateMealCalories()"> Vegetable
Rice(250 cal)</label>

<label><input type="checkbox" value="270"
onchange="updateMealCalories()"> Tomato Rice(270
cal)</label>

<label><input type="checkbox" value="300"
onchange="updateMealCalories()"> Sambar Rice(300
cal)</label>

<label><input type="checkbox" value="220"
onchange="updateMealCalories()"> 3 Boiled
Eggs(220 cal)</label>

<label><input type="checkbox" value="15"
onchange="updateMealCalories()"> 100gms
Cucumber(15 cal)</label>

<label><input type="checkbox" value="40"
onchange="updateMealCalories()"> 40gms Carrot(40
cal)</label>

<label><input type="checkbox" value="420"
onchange="updateMealCalories()"> 250gms
Chicken(420 cal)</label>

</div>

<label><input type="checkbox"
onclick="toggleMeal('snackBox')"> Snacks</label>

<div id="snackBox" style="display:none;">

<label><input type="checkbox" value="160"
onchange="updateMealCalories()"> Peanut(160
cal)</label>


```
<label><input type="checkbox" value="50"
onchange="updateMealCalories()"> Tea(50
cal)</label>
```

```
<label><input type="checkbox" value="80"
onchange="updateMealCalories()"> Fruits(80
cal)</label>
```

```
<label><input type="checkbox" value="300"
onchange="updateMealCalories()"> 4 Bread
Slices(300 cal)</label>
```

```
</div>
```

```
<label><input type="checkbox"
onclick="toggleMeal('dinnerBox')"> Dinner</label>
```

```
<div id="dinnerBox" style="display:none;">
```

```
<label><input type="checkbox" value="200"
onchange="updateMealCalories()"> 2 Chapatti(200
cal )</label>
```

```
<label><input type="checkbox" value="300"
onchange="updateMealCalories()"> Macaroni(300
cal)</label>
```

```
<label><input type="checkbox" value="550"
onchange="updateMealCalories()"> Chicken
Rice(550 cal)</label>
```

```
<label><input type="checkbox" value="420"
onchange="updateMealCalories()"> 150gms
Paneer(420 cal)</label>
</div>
```

```
<input type="number" id="calories"
placeholder="Calories Consumed" readonly>
```

```
<button onclick="calculate()">Generate
Report</button>
```

```
<div id="result"></div>
```

```
<div id="streak"></div>
```

```
<div id="achievements"></div>
```

```
<canvas id="calorieChart"></canvas>
```

```
<h4>  Progress History</h4>
```

```
<canvas id="historyChart"></canvas>
```

```
<button onclick="logout()"
style="background:red;margin-top:10px;">Logout</b
utton>
```

```
</div>
```

```
<script>
```

```
let chart;
```

```
let historyChart;
```

```
// 👁 PASSWORD
```

```
function togglePassword(){
```

```
    let pass=document.getElementById("password");
```

```
    let eye=document.getElementById("eyeIcon");
```

```
    if(pass.type==="password"){
```

```
        pass.type="text";
```

```
        eye.textContent="👁";
```

```
    }else{
```

```
    pass.type="password";  
    eye.textContent="👁️";  
  }  
}
```

// LOGIN

```
function login(){  
  let  
email=document.getElementById("email").value;  
  let  
password=document.getElementById("password").value;  
  
  if(email && password){  
    localStorage.setItem("currentUser", email);  
    showDashboard(email);  
    updateStreak();  
    loadHistoryChart();  
  }else{  
    alert("Enter email & password");  
  }  
}
```

```
}
```

```
// AUTO LOGIN
```

```
window.onload=function(){  
    let user=localStorage.getItem("currentUser");  
    if(user){  
        showDashboard(user);  
        updateStreak();  
        loadHistoryChart();  
    }  
}
```

```
// SHOW DASHBOARD
```

```
function showDashboard(email){  
  
    document.getElementById("authCard").style.display=  
    "none";  
  
    document.getElementById("dashboard").style.display  
    ="block";
```

```
document.getElementById("userEmail").innerText=email;  
}
```

```
// LOGOUT
```

```
function logout(){  
    localStorage.removeItem("currentUser");  
    location.reload();  
}
```

```
// 🔥 STREAK WITH SAME DAY CHECK
```

```
function updateStreak(){  
    let email = localStorage.getItem("currentUser");  
    if(!email) return;  
  
    let streakKey = "streak_" + email;  
    let dateKey = "lastDate_" + email;  
  
    let today = new Date().toDateString();
```

```
let lastDate = localStorage.getItem(dateKey);

let streak =
parseInt(localStorage.getItem(streakKey)) || 0;

if(lastDate === today){
    document.getElementById("streak").innerText =
        " ⚠️ Already logged in today | 🔥 Streak: " +
streak + " days";
    return;
}

streak++;
localStorage.setItem(streakKey, streak);
localStorage.setItem(dateKey, today);

document.getElementById("streak").innerText =
    " 🔥 Daily Streak: " + streak + " days";
}
```

```
// TOGGLE MEAL
```

```
function toggleMeal(id){  
    let box=document.getElementById(id);  
    box.style.display =  
box.style.display=== "none"? "block": "none";  
}
```

```
// CALORIES
```

```
function updateMealCalories(){  
    let total=0;  
  
    let checks=document.querySelectorAll(  
        "#breakfastBox input, #lunchBox input,  
#snackBox input, #dinnerBox input"  
    );  
  
    checks.forEach(ch=>{  
        if(ch.checked){  
            total += parseInt(ch.value);  
        }  
    })  
}
```



```
});
```

```
document.getElementById("calories").value=total;  
}
```

```
// CALCULATE
```

```
function calculate(){
```

```
    let age =
```

```
    parseInt(document.getElementById("age").value);
```

```
    let h =
```

```
    parseInt(document.getElementById("height").value);
```

```
    let w =
```

```
    parseInt(document.getElementById("weight").value);
```

```
    let g = document.getElementById("gender").value;
```

```
    let c =
```

```
    parseInt(document.getElementById("calories").value)
```

```
    || 0;
```

```
    if(!age || !h || !w || !g){
```

```
        alert("Please fill all details");
```

```
        return;
```

```
}
```

```
let bmr = g=== "male"  
  ? (10*w)+(6.25*h)-(5*age)+5  
  : (10*w)+(6.25*h)-(5*age)-161;
```

```
let maintenance = Math.round(bmr*1.55);
```

```
let bmi = (w/((h/100)**2)).toFixed(1);
```

```
document.getElementById("result").innerHTML =  
  " 🔥 Maintenance: "+maintenance+" kcal<br>"+  
  " BMI: "+bmi;
```

```
updateChart(maintenance,c);
```

```
checkAchievements(bmi,c);
```

```
saveHistory(c);
```

```
loadHistoryChart();
```

```
}
```

```
// MAIN CHART
```

```
function updateChart(target,consumed){
    if(chart) chart.destroy();

    chart=new
    Chart(document.getElementById("calorieChart"),{
        type:'bar',
        data:{
            labels:['Target','Consumed'],
            datasets:[{
                label:'Calories',
                data:[target,consumed]
            }]
        }
    });
}
```

// SAVE HISTORY

```
function saveHistory(calories){
    let email = localStorage.getItem("currentUser");
    let key = "history_" + email;
```

```
let today = new Date().toDateString();
```

```
let history =  
JSON.parse(localStorage.getItem(key)) || [];
```

```
let existing = history.find(h => h.date === today);
```

```
if(existing){  
    existing.calories = calories;  
}else{  
    history.push({date: today, calories: calories});  
}
```

```
localStorage.setItem(key, JSON.stringify(history));  
}
```

```
// HISTORY CHART
```

```
function loadHistoryChart(){  
    let email = localStorage.getItem("currentUser");
```

```
let key = "history_" + email;
```

```
let history =  
JSON.parse(localStorage.getItem(key)) || [];
```

```
let labels = history.map(h => h.date);  
let data = history.map(h => h.calories);
```

```
if(historyChart) historyChart.destroy();
```

```
historyChart = new  
Chart(document.getElementById("historyChart"), {  
  type: 'line',  
  data: {  
    labels: labels,  
    datasets: [{  
      label: 'Daily Calories',  
      data: data  
    }]  
  }  
}
```

```
});  
}
```

```
// ACHIEVEMENTS
```

```
function checkAchievements(bmi,cal){
```

```
    let arr=[];
```

```
    if(bmi>=18.5 && bmi<=24.9) arr.push("🏆  
Healthy BMI");
```

```
    if(cal<2000) arr.push("🎯 Calorie Controlled");
```

```
    if(arr.length===0) arr.push("Keep Going 🏃");
```

```
document.getElementById("achievements").innerHTML  
ML =
```

```
    "Achievements:<br>" + arr.join("<br>");
```

```
}
```

```
</script>
```

</body>

</html>

1. Abstract & Introduction

- The Calories Calculator is a web-based application designed to help users calculate their daily calorie intake and understand their health condition using BMI and maintenance calorie calculations. The system allows users to log in, enter personal details such as age, height, weight, and gender, and select their daily meals. Based on this data, the application calculates Body Mass Index (BMI), daily calorie consumption, and maintenance calories.
- The main objective of this project is to provide a simple and user-friendly interface for calculating calorie consumption and maintaining a healthy lifestyle. It also includes charts for better understanding of calorie intake.

2. Architecture Diagram

The application follows a simple client-side architecture:

User (Browser)

↓

HTML (Structure)

↓

CSS (Styling)

↓

JavaScript (Logic & Processing)

↓

Local Storage (Data Storage)

↓

Chart.js (Data Visualization)

3. Description and Working of the Website

The website consists of two main sections:

a) Login Page

- Users enter email and password
- Password visibility toggle is available
- Data is stored in browser local storage

b) Dashboard

After login, the user is redirected to the dashboard where:

- User enters personal details (age, height, weight, gender)
- Selects meals for breakfast, lunch, snacks, and dinner
- Calories are automatically calculated
- System calculates:
 - BMI (Body Mass Index)
 - Maintenance Calories (using BMR formula)

Additional Features:

- Daily Streak Tracking
 - Achievement System
 - Calorie Comparison Chart (Target vs Consumed)
 - History Chart (Daily calorie intake)
-

4. Software Description

Technologies Used:

- HTML: Used for structuring the web pages
- CSS: Used for designing and layout (glassmorphism UI)
- JavaScript: Used for logic, calculations, and interactivity
- Chart.js: Used for graphical representation of data

Features:

- Responsive design
 - Interactive UI
 - Real-time calorie calculation
 - Data visualization using charts
-

5. Backend / Storage Description

This project does not use a traditional backend server.

Storage Used:

- Browser Local Storage

Data Stored:

- User login email
- Daily calorie intake history
- Streak data
- Last login date

Advantages:

- No server required

- Fast performance
- Easy implementation

Limitations:

- Data is not secure
 - Data is lost if browser storage is cleared
 - Not accessible across different devices
-

6. Future Work

- Add real authentication system (Firebase / Backend API)
 - Store data in cloud database
 - Add workout tracking feature
 - Add water intake tracker
 - Mobile app version
 - Personalized diet recommendations using AI
 - Multi-user support with secure login
-

7. Coding

The project is implemented using HTML, CSS, and JavaScript.

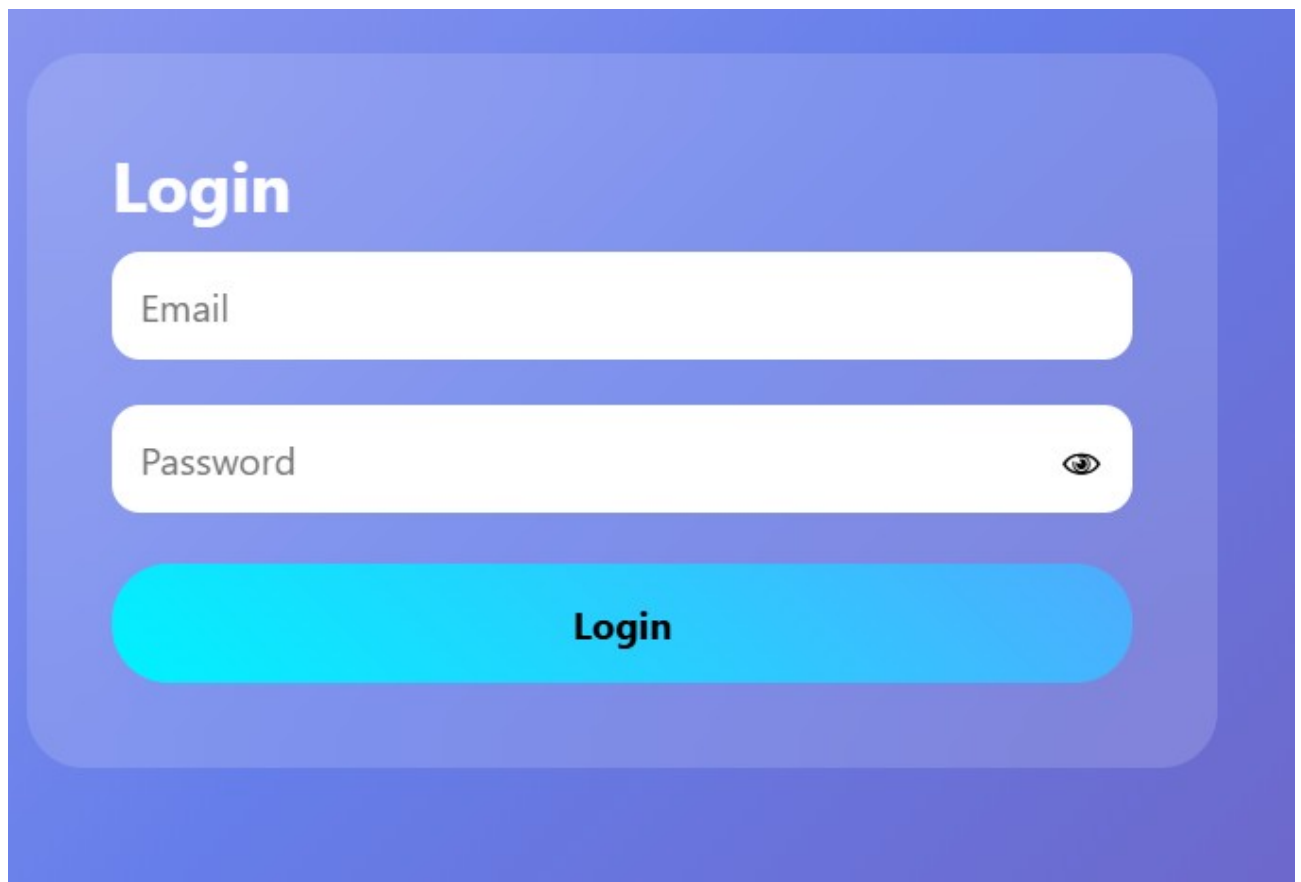
Key Functionalities:

- Login system using local storage
- Password toggle visibility
- Meal selection and calorie calculation
- BMI and BMR calculation

- Chart generation using Chart.js
- Streak and achievement system
- History tracking

8. Screenshots of Your Website

1.Login Page



2. Dashboard Page

Welcome, yamini 🙌

Age

Height (cm)

Weight (kg)

Gender ▼

🍽️ Select Meals

☐

Breakfast

☐

Lunch

☐

Snacks

☐

Dinner

Calories Consumed

Generate Report

2. Meal Selection Section

BREAKFAST

Select Meals

☒

Breakfast

☐

2 Idlis(140 cal)

☐

3 Dosa(360 cal)

☐

Pongal(180 cal)

☐

Oots 1cup(170 cal)

LUNCH

Lunch

☐

Vegetable Rice(250 cal)

☐

Tomato Rice(270 cal)

☐

Sambar Rice(300 cal)

☐

3 Boiled Eggs(220 cal)

☐

100gms Cucumber(15 cal)

☐

40gms Carrort(40 cal)

☐

250gms Chicken(420 cal)

SNACKS

Snacks	☐
Peanut(160 cal)	☐
Tea(50 cal)	☐
Fruits(80 cal)	☐
4 Bread Slices(300 cal)	

DINNER

Dinner	☐
2 Chapatti(200 cal)/label>	☐
Macaroni(300 cal)	☐
Chicken Rice(550 cal)	☐
150gms Paneer(420 cal)	

3.Result Display (BMI & Calories)

Maintenance: 1667 kcal

BMI: 20.0

Daily Streak: 1 days

Achievements:

Healthy BMI

Calorie Controlled

4.Chart Display



5. History Graph



Conclusion

The Calories Calculator project successfully demonstrates how web technologies can be used to build an interactive health monitoring system. It provides useful insights into daily calorie intake and encourages users to maintain healthy habits through visual feedback and tracking features.
